

Fixed Effects Regressions in Corporate Finance

Why specification choice determines what your coefficient means

Dissertation Module

Week 5

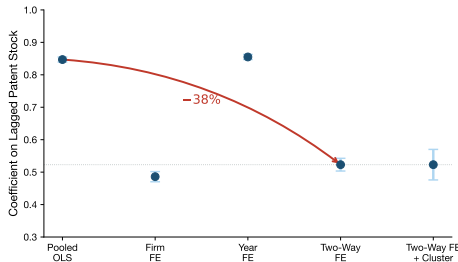
The Problem

Pooled OLS overstates the patent–innovation relationship by 38%

A naive regression of process innovation on patent stock yields $\hat{\beta} = 0.847$.

But this estimate confounds the **within-firm** effect (does a firm innovate more when its patent stock grows?) with the **between-firm** effect (do firms with larger patent stocks innovate more on average?).

The preferred two-way FE specification gives $\hat{\beta} = 0.523$.



Large cross-sectional variation in innovation demands panel methods

Source: Bena, Ortiz-Molina & Simintzi (2022), “Shielding Firm Value: Employment Protection and Process Innovation,” *Journal of Financial Economics*.

	Mean	Std. Dev.	Min	Median	Max
Process innovation	1.137	1.824	0.000	0.000	10.100
Product innovation	1.875	2.193	0.000	0.000	10.479
Lagged patent stock	1.705	1.605	0.000	1.314	9.105
State-year patents	1.671	0.682	0.000	1.712	4.664
Total patent claims	146.2	835.1	0.000	4.000	59,923

45,263

firm-year obs.

4,447

unique firms

1975–1997

sample period

Building the Specification

The same equation under four different assumptions

Pooled OLS — no fixed effects:

$$\text{ProcessInnovation}_{it} = \beta_0 + \beta_1 \text{PatentStock}_{i,t-1} + \beta_2 \text{StateYearPatents}_{it} + \varepsilon_{it}$$

Two-way fixed effects — firm α_i and year γ_t :

$$\text{ProcessInnovation}_{it} = \alpha_i + \gamma_t + \beta_1 \text{PatentStock}_{i,t-1} + \beta_2 \text{StateYearPatents}_{it} + \varepsilon_{it}$$

Stata

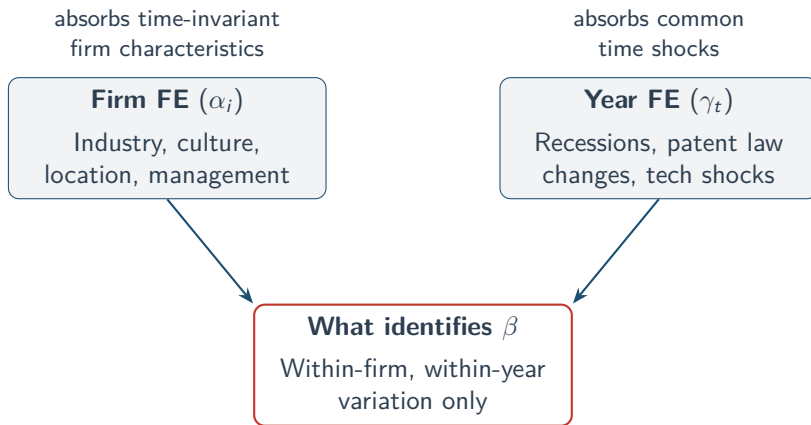
```
reghdfe cl_pcs L1log1pspat  
  logstyrpat,  
  absorb(gvkey fyear)  
  cluster(gvkey)
```

Python (pyfixest)

```
pf.feols(  
  "cl_pcs ~ L1log1pspat  
    + logstyrpat | gvkey + fyear",  
  data=df,  
  vcov={"CRV1": "gvkey"})
```

Note: The original paper clusters by state (treatment level). Here we cluster by firm for illustration.

Firm and year FE together isolate within-firm, within-year variation



Firm FE cuts the patent stock coefficient nearly in half

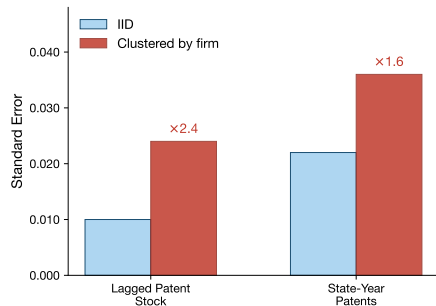
	(1) Pooled OLS	(2) Firm FE	(3) Year FE
Lagged patent stock	0.847*** (0.004)	0.486*** (0.008)	0.855*** (0.004)
State-year patents	−0.019* (0.008)	0.312*** (0.019)	−0.069*** (0.008)
Firm FE		✓	
Year FE			✓
N	45,263	44,898 [†]	45,263
R^2	0.555	0.745	0.569

Column (2): once we compare each firm *to itself* over time, the estimated effect of patent stock on process innovation **drops from 0.847 to 0.486** — much of the pooled estimate reflected persistent cross-firm differences, not a causal within-firm effect.

[†]365 singleton firms (one observation only) are dropped when firm FE are included.

Clustering doubles standard errors but does not change point estimates

	(4) Two-Way FE (IID SE)	(5) Two-Way FE (Clustered SE)
Patent stock	0.523*** (0.010)	0.523*** (0.024)
State-year pat.	0.114*** (0.022)	0.114** (0.036)
Firm + Year FE	✓	✓
S.E. type	IID	Clustered
N	44,898	44,898
R^2	0.749	0.749



IID SEs assume independence. Petersen (2009, *RFS*) shows they are biased downward in panels.

Rule: Cluster at minimum by firm, or at the treatment level.

Putting It Together

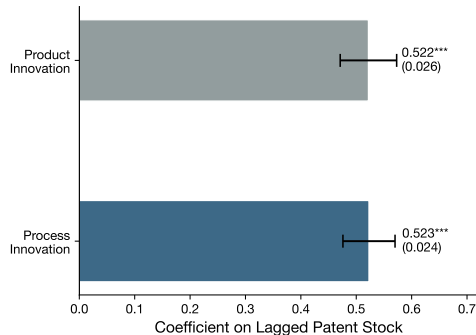
The coefficient halves, but remains significant across all specifications

	(1) Pooled OLS	(2) Firm FE	(3) Year FE	(4) Two-Way FE	(5) Two-Way FE
Lagged patent stock	0.847*** (0.004)	0.486*** (0.008)	0.855*** (0.004)	0.523*** (0.010)	0.523*** (0.024)
State-year patents	-0.019* (0.008)	0.312*** (0.019)	-0.069*** (0.008)	0.114*** (0.022)	0.114** (0.036)
Firm FE		✓		✓	✓
Year FE			✓	✓	✓
S.E. type	IID	IID	IID	IID	Clustered
N	45,263	44,898	45,263	44,898	44,898
R^2	0.555	0.745	0.569	0.749	0.749

Reading left to right: the pooled OLS estimate (0.847) is contaminated by between-firm variation. Firm FE cuts it to 0.486. Adding year FE *on top of* firm FE raises the coefficient slightly to 0.523. Clustering doubles the standard error but the effect remains significant at the 0.1% level.

Patent stock predicts both process and product innovation equally

	Process	Product
Lagged patent stock	0.523*** (0.024)	0.522*** (0.026)
State-year patents	0.114** (0.036)	0.136** (0.042)
Firm FE	✓	✓
Year FE	✓	✓
Clustered SE	✓	✓
<i>N</i>	44,898	44,898
<i>R</i> ²	0.749	0.718



The coefficient on patent stock is nearly identical for both outcomes. The regional innovation environment matters slightly more for product innovation.

For Your Dissertation

Your dissertation regressions need these safeguards

1. **Always include fixed effects.** Pooled OLS produces biased estimates when unobserved heterogeneity correlates with regressors.
2. **Two-way FE (firm + year) is the standard.** Absorbs both persistent firm differences and aggregate time shocks.
3. **Always cluster standard errors.** At minimum by firm. At the treatment level if higher (state, industry).

Show your readers the specification matters

4. **Show multiple specifications.** Present pooled OLS $\rightarrow$ firm FE $\rightarrow$ two-way FE to demonstrate robustness.
5. **Watch how coefficients change.** Large drops signal omitted variable bias in simpler models.
6. **Watch how standard errors change.** Clustering typically inflates SEs — if it doesn't, question whether your panel has meaningful within-cluster correlation.

The same data.

The same regression.

The same variable.

$$\hat{\beta} = 0.847 \quad \text{or} \quad \hat{\beta} = 0.523$$

Fixed effects don't change the data —
they change what variation identifies your estimate.

Data source

Bena, J., Ortiz-Molina, H. & Simintzi, E. (2022). Shielding firm value: Employment protection and process innovation. *Journal of Financial Economics*, 146(2), 637–664.

Methodology

Petersen, M.A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies*, 22(1), 435–480.

Breuer, M., Leuz, C. & Vanhaverbeke, S. (2024). Using and interpreting fixed effects models. *Journal of Accounting Research*, 62(4).

Angrist, J.D. & Pischke, J.-S. (2009). *Mostly Harmless Econometrics*, Chapter 5.